

Pritipriya Dasbehera

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Education

National Institute of Science Education and Research

Bhubaneswar, India

Integrated Masters in Physical Science (INSPIRE Scholar)

October 2021 - May 2026

- Current GPA: **9.49**/10 (University scale), **3.94**/4
- Minor in Computer Sciences
- **Relevant Physics Coursework:**
Advanced Solid State Physics, Quantum Computation and Quantum Information, Many-body physics, Phase Transition and Renormalization, Quantum Field Theory, Non-Linear Dynamics and Chaos, Condensed Matter, Statistical Mechanics, Classical Mechanics
- **Relevant Computer Science:**
Graph Theory and Algorithm Design, Machine Learning, Complexity Theory, Parameterized algorithms
- **Relevant Lab Courses:**
Basic and Advanced Electronics, Optics, Computational, Nuclear Physics, Solid State

Research Articles

Distance by de-correlation: Computing distance with heterogeneous grid cells

Pritipriya Dasbehera, Akshunna S. Dogra, William T. Redman

arXiv, 2025

Research Experience

National Institute of Science Education and Research

Bhubaneswar, India

Thesis | Multipartite entanglement in Many-body Systems

Aug. 2025 - Ongoing

- Supervised by Dr. Anamitra Mukherjee
- Relating one tangle, two tangle and QFI to magnetization and correlation
- Study of entanglement property in 1D XXZ model with longitudinal field

National Institute of Science Education and Research

Bhubaneswar, India

Pre-thesis Project | Quantum Many-body Dynamics

Aug. 2024 - Dec. 2026

- Supervised by Dr. Anamitra Mukherjee
- Low rank inversion (LRI) technique for Greens function computation
- Recursive Green's Function application in Monte-Carlo workflow (**Manuscript under preparation**)

MathePhysics and Johns Hopkins University

Remote

Summer Internship | Biophysics

Jun. 2024 - Nov. 2025

- Supervised by Mr. Akshunna S. Dogra and Dr. William T. Redman
- Distance information by de-correlation based on the variability of grid cell parameters (**Manuscript in arXiv**)

Workshops and Seminars

Asia Pacific QEC Seminar

Remote, Youtube/Discord

Tutorial Series (Attendee) | Quantum Error Correction

20 Aug 2025 - 17 Dec 2025

International Centre for Theoretical Sciences

Bengaluru, India

Discussion Meeting (Attendee) | Geometry, Probability, and Algorithms

12 May 2025 - 16 May 2025

Skills

Programming	Proficient in Python, Julia and C++
Familiar with	Rust, Machine learning frameworks (Python), Web development
Tools	Cluster Computing (Scripting, Scheduling, Parallelization)
Mentorship	Tutoring in undergraduate physics, computer science and robotics

Achievements

2024	Gold medal in Fifth and Sixth semesters , Academic	NISER
2022	Gold medal in First and Second semesters , Academic	NISER
2021	Inspire Scholarship , for 5 years of Int MSc by Department of Science and Technology, Govt. of India	India

University Projects

Machine Learning Renormalisation Group (MLRG)

Bhubaneswar

Phase Transition and Renormalisation [Course] | Term paper

Aug 2024 - Nov 2024

- Supervised by Dr. Sumedha, based on the paper by Hou and You (2023)
- Use of ML to perform RG and construct RG flow without prior knowledge of critical parameters

Anderson Impurity Problem

Bhubaneswar

Many-particle Physics [Course] | Reading project

Aug 2024 - Nov 2024

- Supervised by Dr. Anamitra Mukherjee
- Study of dynamics of electrons in metal doped with impurities
- Different Born approximations and Feynman diagrams
- Group presentation

Secure communication with Chua circuit

Bhubaneswar

Open-Lab (Teaching Lab) | Indigenous experimental setup

Aug 2024 - Nov 2024

- Design and analysis of Chua circuit and study of the non-linear element constructed using op-amp as negative resistance
- Synchronization of two Chua circuits and study of its stability for different parameter values
- Use of synchronization to securely transmit digital and analog information by using chaos to encode information

Co-curricular Experience

National Institute of Science Education and Research

Bhubaneswar, India

RoboTech Club (RTC)

Aug. 2021 - 2024

- **2022:** Organised Student Development Programme (SDP), taught new members Arduino and electronics
- **2023:** Organised Student Development Programme (SDP), Vice president of RTC
- **2024:** Supervised the construction of a 8-bit computer

National Institute of Science Education and Research

Bhubaneswar, India

Informal Teaching Activities

- Undergraduate physics to 1st and 2nd year students
- Python programming to 1st year students

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Outreach Activities

- **2022:** Volunteer in NISER Open Day (outreach for school students)
- **2023:** Conducted the physics session in NISER Open Day

Interests

Robotics Interested in bringing textbook physics and programming to life through robotics

Stargazing Occasionally find myself strolling alone enjoying the night sky

Reading Have always loved reading; thriller, mystery and bleak fiction

Sci. Comm. Regularly promote physics to school students by showcasing interesting experiments and giving theoretical explainers

Languages

English Professional proficiency, [TOEFL 108/120]

Odia Native proficiency

Hindi Full literacy and conversational proficiency